

Final CLIP Credit-Risk Feature-Selection Report

1. Bottom-line verdict

Prompt 8 completes the final analysis and reporting layer from saved artifacts only. The scientific
The current contrastive encoder aligns semantic feature metadata with a limited DEV statistical view

2. Objective and research design

The study compares original statistical selectors, original LLM-assisted selectors, and two frozen C

3. Data, temporal validation, and leakage controls

The final analysis uses saved Prompt 7 OOT predictions and aggregate tables. Home Credit has 120,053

4. Main OOT predictive results

Best fixed-method result by dataset/model:

- homecredit catboost: mrmr AUC 0.7668, Gini 0.5337, PSI 0.0102
- homecredit lr: mrmr AUC 0.7457, Gini 0.4914, PSI 0.0065
- lendingclub_v2 catboost: llm AUC 0.7137, Gini 0.4275, PSI 0.0058
- lendingclub_v2 lr: llm AUC 0.6927, Gini 0.3853, PSI 0.0065

Best CLIP-family result by dataset/model:

- homecredit catboost: best CLIP-family selector 'clip_then_mrmr' AUC 0.7035
- homecredit lr: best CLIP-family selector 'clip_then_mrmr' AUC 0.6632
- lendingclub_v2 catboost: best CLIP-family selector 'clip_then_mrmr' AUC 0.6999
- lendingclub_v2 lr: best CLIP-family selector 'clip_then_mrmr' AUC 0.6842

Across the four dataset/model panels, 'clip_then_mrmr' is consistently stronger than direct 'clip',

5. Score drift, semantic coverage, and redundancy

Model score PSI is taken from saved Prompt 7 run artifacts. DEV score vectors are not persisted, so
Semantic coverage and redundancy are descriptive. Broader semantic coverage does not imply predictiv

6. Representation learning and seed robustness

Prompt 5 retained five seeds and selected seed 55 by the prespecified lowest Home Credit validation

7. LendingClub v2 external validation

LendingClub v2 was external-only. CLIP-family results transfer enough to run valid OOT comparisons,

8. Limitations

The main limitations are recorded in 'results/clip/final_analysis/limitations_register.csv'. The mos

9. Conclusion

CLIP is scientifically usable as an experimental representation and screening extension in this repo